

Jonathan Jacob Koshy

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EDUCATION

University of Waterloo

BASc Electrical Engineering

• **Relevant Coursework:** Signals & Systems, Computer Architecture, Data Struct & Alg, Electronic Circuits, Embedded Systems

Expected Apr. 2029

Waterloo, ON

EXPERIENCE

Firmware Development Intern

Belimo

May 2026 – Aug. 2026

Montreal, QC

- Reduced **PIC18 firmware memory use by 20%** by removing dead code and simplifying memory heavy routines for flash limits
- Diagnosed **CAN, UART, and I2C** timing and framing faults using oscilloscopes and logic analyzers to isolate firmware bugs
- Reduced keypad latency by **30%** by tuning interrupt timing and debounce windows while preserving stable input detection
- Improved gas measurement accuracy by **10%** by filtering **ADC** samples to suppress sensor noise and reject false readings
- Restored **Modbus** communication after malformed frames by correcting UART parsing, frame handling, and state recovery
- Validated firmware against UL test cases across sensor behavior, fault conditions, edge cases, and PCB level operation

Embedded Software Intern

AeroCardia

Sep. 2025 – Dec. 2025

Montreal, QC

- Led core **ESP32 firmware** for a cardiopulmonary wearable, integrating sensor acquisition, BLE telemetry, and OTA updates
- Developed **C drivers** for IMU, PPG, and temperature sensors, enabling synchronized biosignal capture for live monitoring
- Decoupled acquisition from BLE with **FreeRTOS queues** and buffered handoff, reducing dropped biosignal samples by **25%**
- Implemented secure **BLE OTA** with image verification and rollback, cutting wearable firmware update time to under **1 minute**
- Designed an **O2 sensor PCB in Altium** with EEPROM-backed calibration, preserving coefficients across power cycles

Embedded Flight Software Member

Waterloo Aerial Robotics (WARG)

Dec. 2024 – Present

Waterloo, ON

- Implemented **roll/yaw motor mixing** in STM32H7 ZeroPilot, reducing cross-axis coupling in fixed-wing control modes
- Reduced flight oscillation by **20%** by diagnosing attitude-loop instability and retuning **PID gains** during stabilization tests
- Reduced motor-command jitter by **30%** by correcting control-loop timing and actuator scaling in flight-control firmware
- Built a **Quectel EG915Q LTE driver** with UART AT-command FSM, enabling modem control and Raspberry Pi integration

Vehicle Firmware Member

Midnight Sun Solar Car Team

Aug. 2026 – Present

Waterloo, ON

- Improved **cruise-control firmware** by fixing front controller safety FSM faults that block WaveSculptor CAN setpoint tests
- Built a **C FOTA bootloader** that buffers incoming firmware into flash page writes for reliable in field vehicle MCU updates
- Added **CRC32 validation**, sequencing, and ACK retries to reject corrupted or out-of-order firmware before flash writes
- Built reusable **UART/CAN update transport** with ISR-fed circular buffers, decoupling FOTA logic from vehicle bus I/O

PROJECTS

Open Source VR Headset | C, STM32F4, ICM20948, USB HID, OpenVR, KiCad | [Github](#)

- Built a custom **VR tracking stack** for a 2K 120Hz headset, spanning STM32 sensor fusion, USB HID, and SteamVR integration
- Stabilized **6-DoF orientation tracking** with startup IMU bias calibration and Mahony quaternion fusion on STM32F411
- Enabled SteamVR tracking by remapping fused poses into **OpenVR coordinates** and streaming orientation over USB HID
- Added bounded forward tracking with **TF-Luna LiDAR** range baselining, preventing inertial drift in seated movement
- Designed a **2-layer KiCad PCB** with USB-C, regulated power, IMU filtering, I2C routing, and bring-up test points

Edge AI Dementia Companion | C, Python, YOLOv8, Hailo-8L, RPi 5 | [Github](#)

HackCanada Winner

- Built an **edge AI dementia aid** on RPi 5/Hailo-8L that locates misplaced objects, winning Most Likely to Be a Startup
- Offloaded **YOLOv8s** to Hailo-8L, sustaining **1080p 30fps** detection and storing last-seen object locations for real-time search
- Expanded searches beyond camera FOV by driving pan & tilt servos, automatically scanning rooms for missing objects

ESP32 Wi-Fi Media Controller | C++, Python, ESP32, LVGL, TCP/IP, WebSockets | [Github](#)

- Built an **ESP32** touchscreen dashboard using LVGL for PC telemetry, media state, Discord data, and Wi-Fi playback controls
- Designed a queued **TCP/JSON protocol** for bidirectional telemetry and media commands with RGB565 artwork rendering

TECHNICAL SKILLS

Programming: C, C++, Python, RISC-V Assembly, MATLAB, Git

Embedded & OS: FreeRTOS, Linux, ARM Cortex-M, STM32, ESP32, PIC18, Raspberry Pi

Firmware: Bootloaders, OTA/FOTA, ISRs, circular buffers, device drivers, state machines, sensor fusion, control loops

Protocols: SPI, I2C, UART, CAN, BLE, Ethernet, TCP/IP, USB HID, Modbus, PWM, ADC

Hardware & Tools: KiCad, Altium, oscilloscopes, logic analyzers, multimeters, soldering, bench power supplies